LAB 03

Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality

Item	Detail
Topic	Topic 1: Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents
Pipeline stage	Ingestion
Duration	20 min
Simulator page	http://localhost:8000/#/data?tab=instruments&symbol=AAPL
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

The pipeline's research and trading stages must all read the same clean data. The desk's data engineer automates ingestion: bars for the whole universe at four timeframes, a data-quality report that blocks bad files, a cache every later stage reads, and a first unstructured source - news headlines - that will later feed a sentiment agent.

Objective

Ingest bars from Alpaca / Twelve Data / yfinance (or the demo files), run the data-quality checks and score news sentiment.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 03. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 03: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

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Step 1. Fetch data

Data > Instruments & ingestion: fetch AAPL with source 'auto' (Alpaca if keys, else the demo file). Online, try 'yfinance' or 'Twelve Data' too (Twelve Data needs TWELVEDATA_API_KEY).

Open: http://localhost:8000/#/data?tab=instruments&symbol=AAPL (Data > Fetch AAPL)

Check: A toast reports the number of bars, the source and the quality status.

Step 2. Read the data-quality report

Open Data quality for SPY. Look at missing values, duplicate timestamps, gaps over 4 days and big moves (possible unadjusted splits).

Open: http://localhost:8000/#/data?tab=quality&symbol=SPY (Data > Quality SPY)

Check: Status OK or CHECK; the big-move dates include March 2020.

Step 3. Switch timeframes

On the chart, switch SPY between 1Day, 1Week, 1Month and 1Hour.

Open: http://localhost:8000/#/data?tab=chart&symbol=SPY&timeframe=1Week (Data > Chart SPY weekly)

Check: Weekly and monthly bars are resampled from daily bars; hourly bars are US/Eastern.

Step 4. Headlines and sentiment

Data > News & sentiment for NVDA: press 'Fetch & score headlines'. Online with NEWSAPI_KEY it calls NewsAPI.org; otherwise Alpaca News or the synthetic sample headlines. Try the sentiment box with your own sentence.

Open: http://localhost:8000/#/data?tab=news&symbol=NVDA (Data > News NVDA)

Check: Each headline has a VADER score between -1 and +1; the daily sentiment chart appears.

Expected in the app: A cached universe, a data-quality report and scored headlines with daily sentiment.

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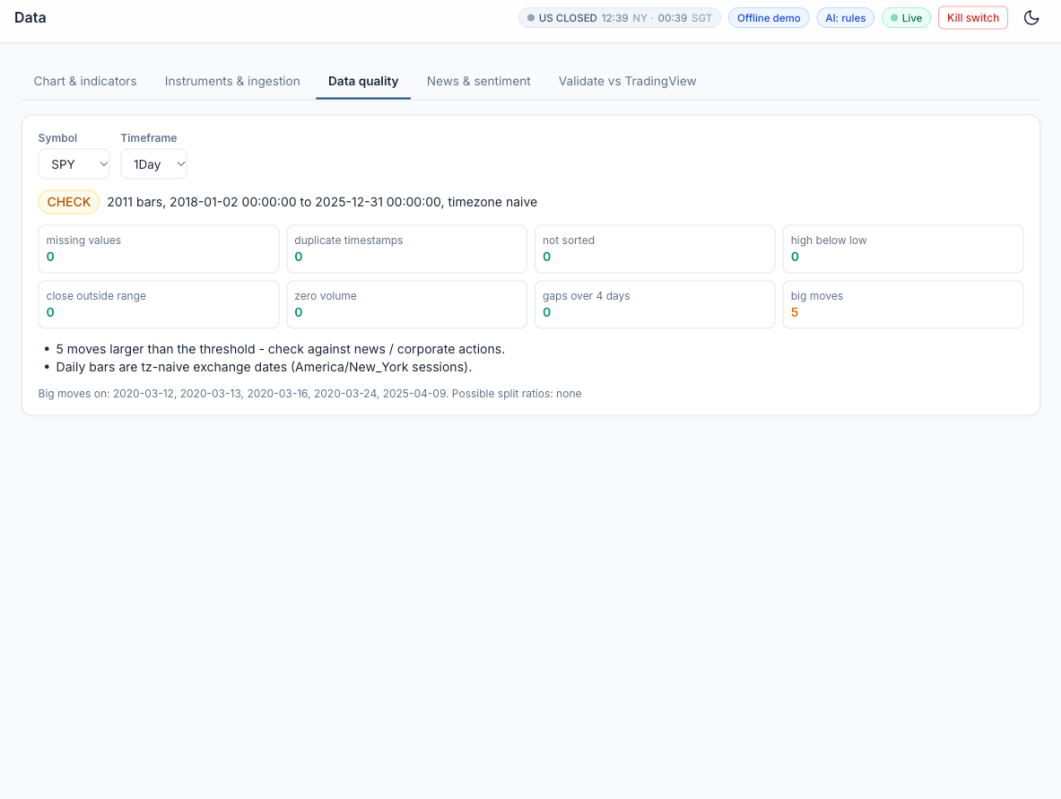


Figure: Lab 03 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

Multi-timeframe data, a data-quality report and headline sentiment - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: download SPY daily bars for 2018-2025 and hourly bars for the last 60 days; resample the daily bars to weekly and monthly; print a data-quality report (missing values, duplicate timestamps, gaps longer than 4 calendar days, daily moves above 15% that may be unadjusted splits). Then get recent NVDA news headlines with yfinance (Ticker('NVDA').news), score each headline with VADER sentiment (vaderSentiment) and print the average sentiment per day.

Open in Claude | Open in ChatGPT | Open Google Colab

Debrief

Ingestion is the first automated stage: it fixes one version of the truth for every later stage and blocks bad data before a model can learn from it. Bars, quotes, trades, news and alternative data are different forms of financial data with different volume, velocity and variety - which is what turns a trading data set into a big-data application.

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